

A Bayesian Hierarchical Spatio-Temporal Analysis of Elevation Effects on Annual Temperature Range in the Contiguous United States

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1 Introduction

Modelling temperature variability across different elevations is important for understanding regional climate systems. Areas at higher elevation are influenced by lower atmospheric pressure and greater snow cover, which can affect annual temperature range [1]. Studying this relationship can help scientists investigate the increased rates of warming observed in high-elevation environments and aid hill farmers in planning crop-growing seasons [2]. Understanding how temperature range varies with elevation is therefore important for characterising regional climate dynamics and the role of topography in climate analyses. This report investigates the relationship between elevation and annual temperature range across the contiguous United States using a Bayesian hierarchical spatio-temporal model. This framework allows us to estimate the effect of elevation while accounting for spatial dependence and inter-annual temporal variation.

2 Data & Methods

2.1 Data Overview

Monthly temperature data from 1901–2024 were obtained from the CRU TS dataset [3]. The analysis was restricted to grid cells within the contiguous United States to retain substantial elevation variation while keeping the dataset computationally manageable. Standardised elevation (metres) for each grid cell was extracted using the `geodata` R package by matching CRU grid centroids to elevation raster data [4,5]. For each spatial location s and year t , the response variable, annual temperature range, was calculated as

$$\text{TempRange}_{st} = \max(\text{Temp}_{st}) - \min(\text{Temp}_{st}), \quad (1)$$

where Temp_{st} denotes the monthly temperatures observed at location s during year t .

2.2 Bayesian Hierarchical Model

2.2.1 Observation Model

Let $Y_{s,t}$ denote the observed annual temperature range at spatial location s and year t . Since temperature range is strictly positive and approximately symmetric on the logarithmic scale, we assume a log-normal observation model,

$$Y_{s,t} \sim \text{LogNormal}(\eta_{s,t}, \sigma_\epsilon^2), \quad (2)$$

where the linear predictor is defined as

$$\eta_{s,t} = \mu + \beta_1 \text{Elev}_s + u_s + \delta(t). \quad (3)$$

Here, μ denotes the overall mean temperature range, Elev_s is the elevation at location s , u_s is a spatial random effect, and $\delta(t)$ represents temporal variation.

2.2.2 Latent Process Model

Spatial effect u_s is modelled using the BYM2 specification, capturing both spatially structured variation and unstructured heterogeneity across locations [6]. The temporal

component $\delta(t)$ is modelled as a second-order random walk (RW2), allowing smooth year-to-year changes in temperature range over the study period [7].

2.2.3 Prior Model

Hyperparameters governing the precision of the spatial effect (τ_u), temporal effect (τ_δ), and observation noise (τ_ϵ) are assigned weakly informative default priors as implemented in R-INLA [8]. Posterior inference is performed using Integrated Nested Laplace Approximation (INLA), which provides deterministic approximations to posterior marginals for latent Gaussian models [9].

3 Results

3.1 Model Selection and Diagnostics

3.1.1 Model Comparison

We compare a sequence of models of increasing complexity using the Widely Applicable Information Criterion (WAIC), a negatively-oriented scoring rule where a lower value indicates better predictive fit [10]. Model M3 corresponds to the full spatio-temporal model defined in Section 2.2.1, while M2 includes elevation with a spatial (BYM2) random effect, M1 includes elevation with a temporal (RW2) random effect, and M0 contains elevation only.

The spatio-temporal model (M3) produces the lowest WAIC value among all candidates (Table 1). This suggests that including both spatial and temporal random effects improves the model's ability to capture variation in annual temperature range across the contiguous United States. Consequently, model M3 is selected for further analysis.

Table 1: Model comparison based on the Widely Applicable Information Criterion (WAIC).

Model	M3	M2	M1	M0
WAIC	1,803,514	1,972,100	2,875,434	2,892,597

3.1.2 Evaluation of Selected Model

M3 was evaluated using probabilistic goodness-of-fit measures, including posterior predictive p-values (PPP) and the continuous ranked probability score (CRPS). PPP values near 0.5 indicate adequate model fit [11]. M3's PPP values of 0.68 (mean) and 0.72 (variance) (Table 2) suggest that the model adequately captures both mean and variance, but with slight overestimation. Table 2 also displays M3's CRPS, a negatively-oriented score measuring how well the model predicts the observed data [12]. The CRPS of 1.0561°C is small relative to the observed temperature range (3.9 – 49.2°C), indicating that the model provides accurate probabilistic predictions and effectively captures the variability in the data.

Table 2: Continuous Ranked Probability Score (CRPS) and Bayesian Posterior Predictive p-Values (PPP) of M3.

Model	CRPS	Mean PPP	Variance PPP
M3	1.0561	0.6800	0.7200

The model’s assumption of log-normality is assessed using a Quantile–Quantile (Q–Q) plot of the Bayesian studentised residuals (Figure 1).

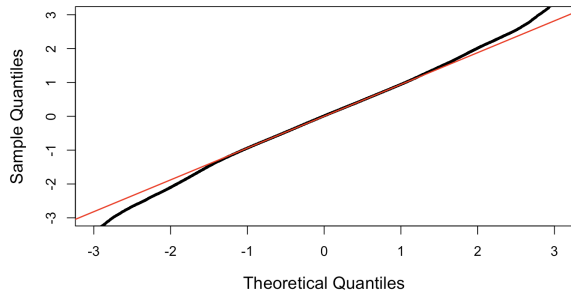


Figure 1: Quantile-Quantile (Q–Q) plot of Bayesian studentised residuals showing residuals (black points) versus theoretical quantiles (red line) under the log-normal model.

Figure 1 shows that the residuals closely follow the theoretical quantiles expected under the log-normal model, indicating that the log-normality assumption is sufficiently satisfied. Minor deviations in the tails suggest slightly heavier tails than expected, though the majority of the distribution aligns with model expectations.

3.2 Effect of Elevation on Annual Temperature Range in the Contiguous United States

Elevation exhibits a positive association with annual temperature range, with a posterior mean estimate of 0.0390 and a 95% credible interval of [0.0271, 0.0510] (Table 3). Since the credible interval excludes zero, there is strong evidence that elevation is positively associated with annual temperature range.

Table 3: Posterior estimates (mean, standard deviation (SD), and lower and upper bounds of 95% credible intervals) of fixed effects from the Bayesian spatio-temporal model.

Covariate	Posterior Mean	SD	2.5% CI	97.5% CI
Intercept	3.2008	0.0037	3.1935	3.2081
Elevation	0.0390	0.0061	0.0271	0.0510

At average elevation and under mean spatial and temporal conditions, the expected annual temperature range is $\exp(3.2008) \approx 24.6^\circ\text{C}$. Furthermore, holding spatial and temporal effects constant, a one-unit increase in standardised elevation (approximately 702 metres) is associated with roughly a 4% increase in annual temperature range ($\exp(0.0390) \approx 1.04$). Although elevation is spatially structured, this relationship persists after accounting for spatial dependence through the BYM2 random effect, indicating that elevation explains variation beyond broad spatial patterns.

3.3 Contribution of Spatial and Temporal Effects to Temperature Range Variability

In addition to elevation, the model incorporates spatial and temporal random effects to capture residual variation in annual temperature ranges. The temporal effect captures year-to-year deviations shared across locations. Its estimated variance of 0.014 suggests relatively modest inter-annual variation in temperature range from 1901 to 2024. The spatial component represents location-specific deviations not explained by

elevation or temporal effects. The estimated mixing parameter from the BYM2 model ($\phi = 0.12$) indicates that only a small proportion of spatial variability is spatially structured, while the majority arises from unstructured local variation.

Figure 2 shows that major mountainous regions, such as the Rockies and Appalachians, exhibit negative spatial effects. However, the overall pattern does not closely follow elevation, suggesting the spatial effect captures residual spatial variability not explained by elevation alone.

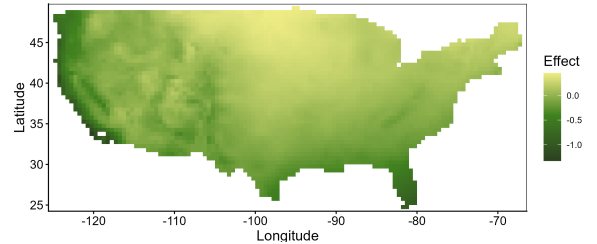


Figure 2: Spatial random effect of model M3 across the contiguous United States.

4 Discussion

The results provide clear evidence that elevation exerts a meaningful positive effect on annual temperature range across the contiguous United States. In particular, the model suggests that a one-standard deviation increase in elevation (approximately 702 metres) is associated with roughly a 4% increase in annual temperature range. This finding is consistent with the expectation that higher-elevation regions experience greater temperature variability due to reduced atmospheric pressure and reduced thermal buffering. The superior performance of the selected spatio-temporal model (M3) highlights the presence of both spatial and temporal variability in temperature range. However, the estimated temporal variance of 0.014 is relatively small, and the spatial mixing parameter ($\phi = 0.12$) indicates that only a small proportion of spatial variation is structured, suggesting that most spatial variability occurs at more local scales.

Several limitations of the modelling approach should be acknowledged. The Q–Q plot of studentised residuals indicates a reasonable assumption of log-normality, but due to slight tail deviation, predictions in climatologically extreme regions may not be as accurate as those in more typical elevation ranges. Additionally, the data consisted of monthly temperature averages, limiting the model’s ability to capture finer temporal variability that may be important in understanding local climate dynamics.

Future work could address these limitations by incorporating additional environmental covariates, such as proximity to large water bodies or land cover type, to help explain remaining spatial heterogeneity. Alternative spatial modelling approaches, such as nearest-neighbour or low-rank Gaussian process approximations, could capture finer-scale spatial structure while remaining computationally feasible [13]. Finally, extending the model to include a space–time interaction would enable location-specific temporal trends, potentially improving both model fit and interpretability.

Use of Generative AI

We acknowledge the use of generative artificial intelligence to assist with concept clarification and grammar checking.

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Appendix A Data Preparation and Raster Processing

This appendix provides the main steps used to preprocess the CRU temperature dataset and prepare the spatial data used in the analysis. The preprocessing includes cropping the dataset to the contiguous United States, aligning coordinate systems, and generating elevation covariates.

```
# Open CRU dataset
df <- rast("cru_ts4.09.1901.2024.tmp.dat.nc", subds = "tmp")

# Load US states shapefile
us <- st_read("cb_2018_us_state_500k/cb_2018_us_state_500k.shp")

# Ensure CRS is lon/lat
if (is.na(st_crs(us))) {
  st_crs(us) <- 4326
}

us <- st_transform(us, 4326)

# Remove Alaska, Hawaii, and territories
drop <- c("AK", "HI", "PR", "AS", "VI", "GU", "MP")
us <- us %>% filter(!(STUSPS %in% drop))

# Dissolve into single CONUS polygon
us_union <- st_union(us)

# Adjust spatial grid covariates to temperature anomalies (same CRS & res)
# Ensure elevation has the same coordinate reference system as the anomaly grid
# (use ref)
# Reproject to match the reference raster if it's different
if (!same.crs(r_elev, ref)) r_elev <- project(r_elev, ref)

# Resample elevation so that its resolution and grid alignment
r_elev <- resample(r_elev, ref, method = "bilinear")

# get only single elevation layer
r_elev <- r_elev[[1]]

# Create a time index and label layers
# Build a monthly date sequence starting from '1961-01-01'
# Assign layer names in 'YYYY-MM' format for consistent reshaping
# Subset layers by year range using 'keep_layers'

full_start <- as.Date("1901-01-01")
tt_full <- seq(full_start, by="month", length.out = nlyr(r_temp))
keep_layers <- which(tt_full >= as.Date("1901-01-01"))
r_temp <- r_temp[[keep_layers]]
tt <- tt_full[keep_layers]
names(r_temp) <- format(tt, "%Y-%m")

# Reshape data from raster to long format (cell × year)
- Convert raster to a wide data frame with 'as.data.frame(..., cells=TRUE, xy=TRUE)'.
- Convert to long format using 'melt()' into columns:
- spatial identifiers ('cell', 'lon', 'lat')
- time identifier ('ym')
- response ('tmp_range')

# convert r_temp to season temperature ranges
df_temp <- as.data.frame(r_temp, xy = TRUE, cells = TRUE, na.rm = FALSE)
```

```
dt <- melt(as.data.table(df_temp),
           id.vars = c("cell","x","y"),
           variable.name = "ym",
           value.name = "tmp")
dt[, year := as.integer(substr(ym, 1, 4))]
dt[, month := as.integer(substr(ym, 6, 7))]
dt <- dt[!is.na(tmp)]

dt_range <- dt[, .(tmp_range = max(tmp, na.rm = TRUE) - min(tmp, na.rm = TRUE)),
               by = .(x, y, cell, year)]
setnames(dt_range, c("x","y"), c("lon","lat"))

# Merge static covariates (robust join by coordinates)
- Convert elevation rasters to '(x,y,value)' tables.
- Round coordinates to avoid floating-point mismatch. Coordinates were rounded to six decimal places
in order to avoid floating-point precision issues during spatial joins. References : Pebesma, E. (2018).
Simple Features for R: Standardized Support for Spatial Vector Data. R Journal
- Merge covariates into the long table using '(lon, lat)' join (not 'cell').
- Scale covariates:
  - 'elev_sc = scale(elev)'

df_elev <- as.data.frame(r_elev, xy = TRUE, na.rm = FALSE)
df_elev <- as.data.table(df_elev)

setnames(df_elev, names(df_elev)[3], "elev")

# round coords to avoid floating-point join issues
df_elev[, 'x' := round(x, 6), y := round(y, 6)]
dt_range[, 'lon' := round(lon, 6), lat := round(lat, 6)]

# merge covariates then join to dt_range
dt_range <- merge(
  dt_range,
  df_elev,
  by.x = c("lon","lat"),
  by.y = c("x","y"),
  all.x = TRUE
)

# drop rows missing covariates (avoid silent drops inside INLA)
dt_range <- dt_range[!is.na(elev)]

# scale covariates
dt_range[, elev_sc := as.numeric(scale(elev))]

# reindex spatial after filtering
stopifnot(nrow(dt_range) > 0)
dt_range[, area_id := as.integer(factor(cell))]
dt_range[, cell := NULL]
dt_range$t_id <- as.integer(factor(dt_range$year))

## Create spatial index
Define spatial ID: 'area_id = factor(cell)' after filtering

dt_inla <- copy(dt_range)
S <- max(dt_inla$area_id)
```

Appendix B Construction of Spatial Neighbourhood Graph

The spatial adjacency structure required for the BYM2 model was constructed using rook adjacency between neighbouring grid cells. The adjacency matrix was converted into an INLA graph format.

```
#Arrange cells on the grid using unique 'lon/lat' #Build rook adjacency using 'cell2nb()'
#Convert to an INLA graph file and read it with 'inla.read.graph()'

```{r}
cells <- unique(dt_inla[, .(area_id, lon, lat)])
setorder(cells, lat, lon)

ux <- sort(unique(cells$lon))
uy <- sort(unique(cells$lat))
ncol <- length(ux)
nrow <- length(uy)

cells[, col := match(lon, ux)]
cells[, row := match(lat, uy)]

mat <- matrix(NA_integer_, nrow = nrow, ncol = ncol)
mat[cbind(cells$row, cells$col)] <- cells$area_id
idx_to_area <- as.vector(mat)

nb_full <- cell2nb(nrow, ncol, type = "rook")
nb_area <- vector("list", length = S)

for (idx in seq_along(nb_full)) {
 a <- idx_to_area[idx]
 if (is.na(a)) next
 neigh_idx <- nb_full[[idx]]
 neigh_area <- idx_to_area[neigh_idx]
 neigh_area <- neigh_area[!is.na(neigh_area)]
 nb_area[[a]] <- sort(unique(as.integer(neigh_area)))
}

for (i in seq_len(S)) if (is.null(nb_area[[i]])) nb_area[[i]] <- integer(0)

write_inla_graph <- function(nb, file) {
 S <- length(nb)
 con <- file(file, open = "wt")
 on.exit(close(con), add = TRUE)
 writeLines(as.character(S), con)
 for (i in seq_len(S)) {
 neigh <- nb[[i]]
 if (is.null(neigh)) neigh <- integer(0)
 neigh <- sort(unique(as.integer(neigh)))
 writeLines(paste(c(i, length(neigh), neigh), collapse = " "), con)
 }
}

graph_file <- file.path(tempdir(), "usa_grid.graph")
write_inla_graph(nb_area, graph_file)
g <- inla.read.graph(graph_file)
```

## Appendix C INLA Model Implementation and Diagnostic Checks

This appendix presents the R-INLA model specifications used to estimate the Bayesian spatio-temporal model and the diagnostic procedures used to assess model fit.

```
ctrl_pred <- list(compute = TRUE, link = 1)

ctrl_comp <- list(
 dic = TRUE,
 waic = TRUE,
 cpo = FALSE,
 config=TRUE
)

inla.setOption(num.threads = 8)

M0: fixed effects only (no spatial/temporal structure)
m0 <- inla(
 tmp_range ~ 1 + elev_sc,
 data = dt_inla,
 family = "lognormal",
 control.predictor = ctrl_pred,
 control.compute = ctrl_comp
)

M1: + temporal RW2 trend
m1 <- inla(
 tmp_range ~ 1 + elev_sc +
 f(t_id, model = "rw2", constr = TRUE),
 data = dt_inla,
 family = "lognormal",
 control.predictor = ctrl_pred,
 control.compute = ctrl_comp
)

M2: + spatial BYM2 effect
m2 <- inla(
 tmp_range ~ 1 + elev_sc +
 f(area_id, model = "bym2", graph = g,
 scale.model = TRUE, constr = TRUE),
 data = dt_inla,
 family = "lognormal",
 # control.fixed = ctrl_fxd,
 control.predictor = ctrl_pred,
 control.compute = ctrl_comp
)

M3: full spatio-temporal model
m3 <- inla(
 tmp_range ~ 1 + elev_sc +
 f(t_id, model = "rw2", constr = TRUE) +
 f(area_id, model = "bym2", graph = g,
 scale.model = TRUE, constr = TRUE),
 data = dt_inla,
 family = "lognormal",
 # control.fixed = ctrl_fxd,
 control.predictor = ctrl_pred,
 control.compute = ctrl_comp
)
```

```
Model Comparison
cat("M0 WAIC =", m0$waic$waic, "\n")
cat("M1 WAIC =", m1$waic$waic, "\n")
cat("M2 WAIC =", m2$waic$waic, "\n")
cat("M3 WAIC =", m3$waic$waic)

Model summaries
m3$summary.fixed
m3$summary.hyperpar

Posterior predictive checks:
sample for model
n_sims <- 100
y_rep <- matrix(NA, nrow = nrow(dt_inla), ncol = n_sims)
post_samples <- inla.posterior.sample(n_sims, m3)
for(i in 1:n_sims){
 s <- post_samples[[i]]
 eta <- s$latent[1:nrow(dt_inla)]
 sigma <- 1 / sqrt(s$hyperpar[[1]])
 y_rep[,i] <- rlnorm(nrow(dt_inla), meanlog = eta, sdlog = sigma)
}
y_rep_all <- as.vector(y_rep)

Residual diagnostics using bayesian studentized residuals:
compute residuals
n <- nrow(dt_range)
y <- log(dt_range$tmp_range)
X <- cbind(1, dt_range$elev_sc)
XtX_inv <- solve(t(X) %*% X)
h <- rowSums((X %*% XtX_inv) * X)

studentisedred <- matrix(NA, nrow = n, ncol = n_sims)
for(l in 1:n_sims){
 s <- post_samples[[l]]
 eta <- s$latent[1:n]
 sigma <- 1 / sqrt(s$hyperpar[[1]])
 studentisedred[,l] <- (y - eta) / sqrt(sigma^2 * (1 - h))
}
studentisedredm <- apply(studentisedred, 1, mean)

Normality (qq plot) - checks log normality of data
qqnorm(studentisedredm,
 main="",
 xlab = "Theoretical Quantiles",
 ylab = "Sample Quantiles",
 pch = 16, cex = 0.4, col = "black",
 xlim = c(-3,3), ylim = c(-3,3),
 cex.lab = 1.25)
qqline(studentisedredm, col = "red", lwd = 1.5)

More numeric absolute measures of fit
Bayesian posterior predictive p-values of mean and variance

yrepmean <- apply(y_rep, 2, mean)
yrepvar <- apply(y_rep, 2, var)
obs_mean <- mean(dt_inla$tmp_range)
```



```
obs_var <- var(dt_inla$tmp_range)

p_mean <- mean(yrepmean >= obs_mean)
p_var <- mean(yrepvar >= obs_var)

bayes_pvals <- c(
 mean = p_mean,
 variance = p_var)

print(bayes_pvals)

Calculate CRPS
crps_values <- crps_sample(y = dt_range$tmp_range, dat = y_rep)
crps_mean <- mean(crps_values)
cat("CRPS =", round(crps_mean, 3), "\n")
```